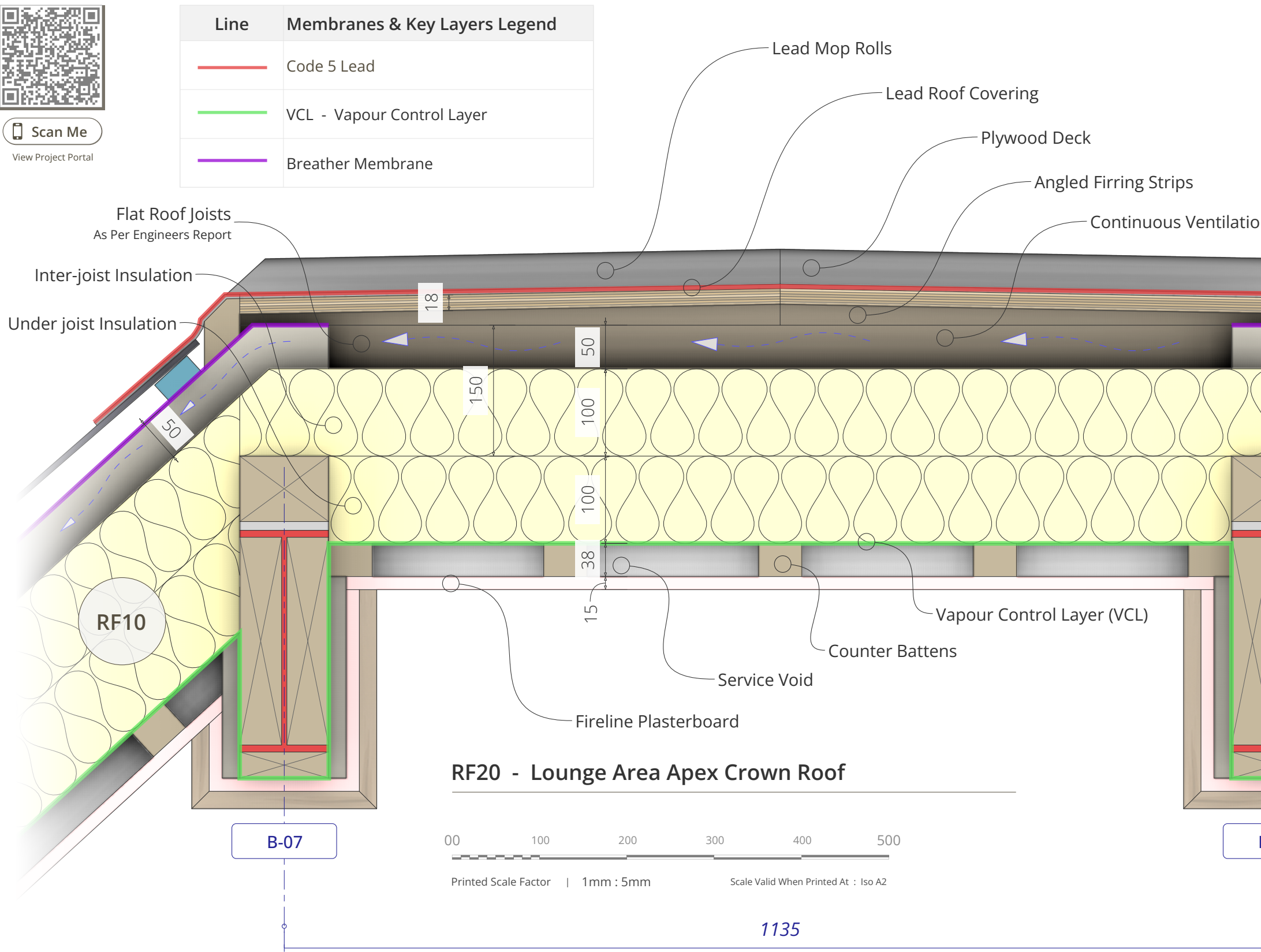




Calculation			
U = 1 / (0.04 + 0.00 + 0.00 + 0.00 + 0.00 + 3.01 + 4.55 + 0.00 + 0.48 + 0.05 + 0.01 + 0.10)			
= 1 / 8.24 = 0.12 W/m²K			
Layer	Thickness (mm)	Lambda (W/mK)	R-value (m²K/W)
External Surface Resistance (Rse)	-	-	0.040
Roof Covering - Code 5 Lead (Milled)	2.24	-	0.000
Separation Layer - Building Paper/Geotextile	-	-	0.000
Sub-Deck - 18mm Structural Plywood (BS EN 636-3)	18	-	0.000
Furring Strips - Tapered Softwood	-	-	0.000
Ventilated Air Gap - Top 50mm of Joist Zone + Furring Zone	50	-	0.000
Insulation - Inter-joist PIR (Bridged 10.4% by Joists)	100	0.028	3.005
Insulation - Continuous Under-joist PIR (GA4000)	100	0.022	4.545
VCL - Virgin Polythene (1000g)	0.25	-	0.000
Service Void - 38mm Batten Cavity (Bridged 12.5%)	38	0.078	0.478
Internal Lining - Gyproc FireLine (12.5mm)	12.5	0.250	0.050
Internal Finish - Thistle Multi-Finish Skim	3	0.570	0.005
Internal Surface Resistance (Rsi) - Upward Flow	-	-	0.100
Total R-value			8.223
Result			
U-Value			0.12 W/m²K

Value Compliant - England Part L1B : Roof U-value 0.15 W/m²K.

Element	Specifications
Lead Roof Covering	Supply and install Milled Lead Sheet Code 5 to BS EN 12588. Lay in bays not exceeding 600mm width (between rolls) and 2250mm length. Fixings: Copper ring-shank nails (3.35mm gauge) staggered at 50mm centres at the head. Joints: Wood-cored rolls (45mm height) for longitudinal joints. Treatment: Apply patination oil immediately after installation to prevent carbonate runoff staining adjacent slates.
Separation Layer	Install Building Paper (Class A to BS 1521) or a dedicated Geotextile Lead Underlay fleece. Lay with 150mm overlaps. Function: This is CRITICAL to isolate the lead from the plywood deck. Direct contact between lead and timber resins/glues will cause chemical corrosion from the underside.
Sub-Deck	Construct deck from 18mm Structural Plywood (to BS EN 636-3 Service Class 3). Fix to firrings using corrosion-resistant screws. Expansion Gap: Leave a 3mm gap between boards and a 10mm gap at all perimeters to allow for moisture movement. Note: OSB is NOT permitted for lead work.
Furring Strips	Install preservative-treated softwood firrings on top of joists to create a minimum finished fall of 1:60 (Design to 1:40 recommended to account for deflection). Configuration: Crown Roof falling from centre ridge to eaves.
Structural Joists	Install 47mm x 150mm C24 Treated Softwood Joists at 450mm centres. CRITICAL: Joists must bear onto wall plates/hangers sitting ABOVE the steel beams. If not possible, joists must be notched but ensure the top of the joist is a minimum of 50mm CLEAR of the steel beam top flange. This creates the essential continuous air path from the pitched roof through this flat roof.
Ventilation Void	Maintain a guaranteed continuous clear air path of 50mm between the underside of the deck/furring and the top of the insulation. Ensure this path is not blocked by solid blocking, headers, or the steel beams.
Inter-Joist Insulation	Friction fit 100mm Celotex GA4000 (Rigid PIR) between joists. Boards must be cut accurately to fill the void without gaps. Ensure insulation does not rise into the 50mm ventilation void (use batten clips if necessary).
Under Joist Insulation	Install a continuous layer of 100mm Celotex GA4000 (or equivalent PIR) to the underside of the joists. Joints must be tightly butted. This layer eliminates thermal bridging through the timber structure.
Vapour Control Layer (VCL)	Supply and install a dedicated 1000-gauge (250 micron) virgin polythene Vapour Control Layer (e.g., Visqueen High Performance VCL) over the warm side of the continuous PIR insulation. This must be installed BEFORE the counter-battens. Lap all joints by minimum 150mm and seal with double-sided butyl tape. Seal perimeter to walls/steels with acoustic sealant.
Counter-Battens	Install 38mm x 50mm treated softwood battens at 400mm centres perpendicular to joists. These battens clamp the VCL and Insulation to the joists. The 38mm depth creates the service void for cabling. Warning: Ensure batten thickness accommodates any proposed recessed lighting depth.
Structural Fixings (CRITICAL)	Fix the Counter-Battens through the VCL and 100mm Under-Joist PIR into the timber joists using specialised Structural Insulation Screws (e.g., Timco Insulation Fixings, EJOT Dabo). Total fixing length approx 200mm. While shear load is lower on flat ceilings, these specific fixings are required to support the dead load of the 100mm insulation preventing it from sagging over time. Standard screws are not sufficient.
Internal Lining	Fix 12.5mm Gyproc FireLine (or Knauf Fire Panel) to the counter-battens. Screw fixings must penetrate the timber batten but NOT puncture the VCL behind it. Stagger joints.
Internal Finish	Apply 3mm Thistle Multi-Finish skim coat to a smooth finish.
Design Rationale	We have elected for a Code 5 Lead Crown Roof to minimize vertical profile height. The lead acts as both covering and flashing. The steel beams have been dropped relative to the roof joists to allow the 50mm ventilation path to pass uninterrupted over the steelwork, eliminating the risk of condensation pooling in a 'cold roof' dead spot. The U-value target matches the high-performance RF10 specification (approx 0.12 W/m²K).
Important Considerations	1. CHIMNEY HEIGHT: The adjacent chimney pot/terminal MUST terminate a minimum of 650mm above the finished level of this flat roof to satisfy Part J. 2. FIRE BREAK: The timber deck and Lead must stop 50mm short of the chimney masonry, separated by non-combustible rockwool. 3. STEELWORK: If steel beams are to be clad in Oak, the 12.5mm FireLine must encase the steel/void first to maintain structural fire protection before the aesthetic oak is fitted.
Standards	BS EN 12588 (Lead), BS 6915 (Lead Work Design), BS 5250 (Control of Condensation), Approved Document L (2025), Approved Document B (Fire Safety), BS EN 636-3 (Plywood).